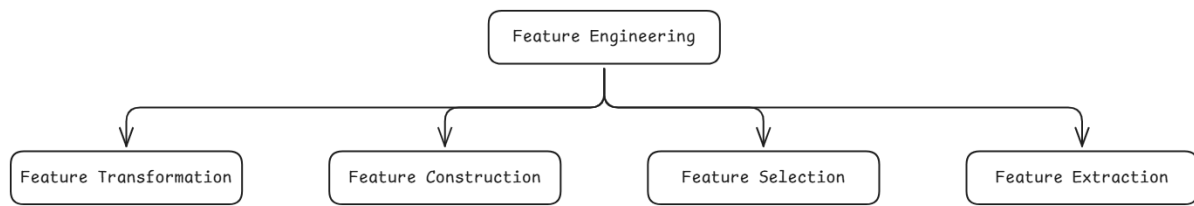


EDA Revision Session

Agenda

- Handling Missing Data
- Detecting and Handling Outliers
- Standardizing Inconsistent Text Data
- Data Transformation Techniques
- Normalisation and Standardization
- Categorical Data Encoding
- Data Aggregation
- Creating and Interpreting Matplotlib Visualizations

[Link to Notebook](#)



Feature engineering is the process of creating, modifying, selecting, and preparing input variables (features) so that a machine learning model can learn patterns more effectively.

A good way to think about it is:

Raw data → Feature Engineering → Better Features → Better Model Performance

Feature engineering is often one of the most impactful parts of building machine learning systems.

Main Categories of Feature Engineering

These four categories are the major ones:

1. Feature Transformation
2. Feature Construction
3. Feature Selection
4. Feature Extraction

There are also a few additional categories commonly included in practice, such as:

5. Feature Encoding
6. Feature Scaling
7. Feature Generation / Augmentation
8. Feature Reduction (sometimes considered part of extraction)
9. Handling Missing Values and Outliers (often grouped under feature preprocessing)

F1. Feature transformation converts an existing feature into a different format while keeping the same information.

The goal is usually to make relationships easier for models to learn.

Examples

Original Feature: Income = \$100,000

Transformations:

- $\text{Log}(\text{Income})$
- $\text{Square Root}(\text{Income})$
- Income^2
- Standardized Income
- Normalized Income

Another example:

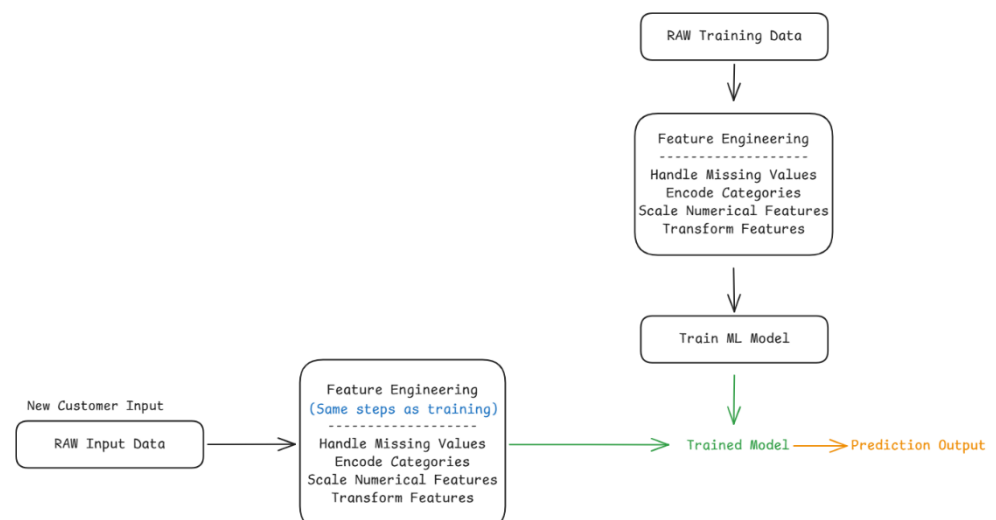
- Age = 35

Transform into:

- Age Category = Adult

Common transformations

- Log transformation
- Square root
- Reciprocal
- Power transformation
- Binning
- Scaling
- Normalization



F2. Feature construction creates **new features** by combining existing ones or using domain knowledge. [\[Add More Information\]](#)

This is sometimes called **feature creation**.

Height	Weight
180 cm	80 kg

Construct: $BMI = Weight / Height^2$

BMI did not exist originally.

Another example

Original:

- Purchase Date
- Delivery Date

Construct:

- Delivery Time = Delivery Date – Purchase Date

Another example

- Length
- Width

Construct:

- Area = Length × Width

Another example

Debt	Income
\$20,000	\$80,000

- Feature Construction: Debt-to-Income Ratio = Debt / Income
- Calculation: $20,000 / 80,000 = 0.25$ or 25%

New Dataset:

Debt	Income	Debt-to-Income Ratio
20,000	80,000	25%

F3. Feature selection chooses the most useful features while removing irrelevant or redundant ones.

Unlike feature extraction, the original features remain unchanged.

Suppose you have:	Feature Selection
• Age • Gender • Income • Zip Code • Customer ID • Random Noise	Feature selection may keep: • Age • Income • Gender And remove: • Customer ID • Random Noise

Benefits

- Faster training
- Less overfitting
- Better interpretability
- Lower storage

Methods

Filter Methods: Independent of machine learning model.

Examples

- Correlation
- Chi-square
- Mutual Information
- ANOVA

Wrapper Methods: Evaluate subsets using a model.

Examples

- Forward Selection
- Backward Elimination
- Recursive Feature Elimination (RFE)

Embedded Methods: Selection occurs during model training.

Examples

- LASSO
- Decision Trees
- Random Forest
- Gradient Boosting

F4. Feature extraction creates **new lower-dimensional features** from existing ones.

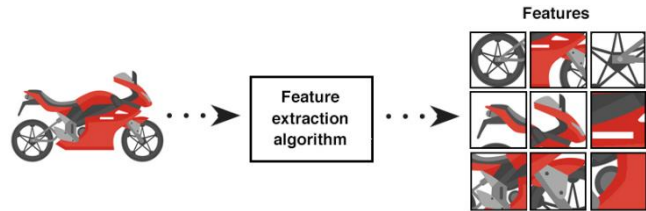
The original features are transformed into a new feature space. [Doesn't Add New Info]

Unlike feature construction, these new features often have no obvious physical meaning.

Example

Original

- 1000 features

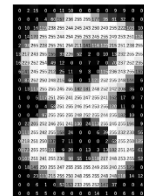


After extraction

- 20 principal components

Examples

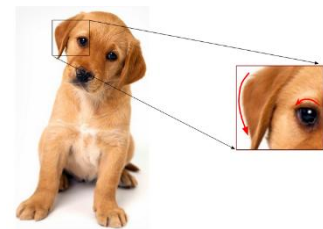
- PCA
- LDA
- Autoencoders
- t-SNE (primarily visualization)
- UMAP (primarily visualization)
- ICA
- Word Embeddings
- CNN feature embeddings



```

0 0 25 11 11 31 0 0 0 0 0 0 0 0 0 0
0 0 0 0 4 00 37 30 20 20 20 137 7 40 31 32 0 29
0 10 19 219 208 208 240 240 240 210 210 250 222 203 31
0 14 170 255 255 244 254 254 254 254 254 254 254 124 1
2 0 90 208 208 208 208 210 210 210 132 132 215 208 235 49
13 17 13 208 208 165 126 26 82 2 0 2 13 82 208 208 49
19 259 262 256 49 12 0 0 0 0 0 7 7 237 257 236 32
41 45 10 208 208 237 18 0 0 0 0 7 7 111 7 208 256 132
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 13 13 208 208 208 208 208 208 208 208 208 208 208 30
1 0 1 517 215 208 208 208 208 208 208 208 208 208 208 208
0 0 0 4 58 255 255 255 254 254 255 255 255 121 0
0 0 9 9 275 255 255 254 254 254 255 255 254 182 10 0
0 22 294 252 254 244 244 244 130 130 255 254 254 254 0
0 1111 255 255 265 264 18 0 0 0 0 3 9 95 352 30 56
0 218 136 208 132 7 11 0 0 0 0 3 9 95 265 300 135
1 97 255 255 91 9 26 0 2 2 13 82 255 245 45 0
0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
0 18 184 255 255 255 255 255 255 255 255 255 255 128
2 13 21 23 255 255 255 249 255 255 249 249 134 12 0
0 0 4 1 0 120 239 255 252 247 37 20 0 0 1
0 0 5 5 0 0 0 0 0 1 14 1 6 0 0 0

```



Example

Original

- Height
- Weight
- Age
- Income
- Education
- ...

Extract

- PC1
- PC2
- PC3

These principal components summarize most of the information.

Data preprocessing is the process of cleaning, transforming, and preparing **raw data** before analysis or machine learning.

1. Handling Missing Data

Missing values occur when some observations have no recorded value.

Example

Age	Salary
25	50000
30	NaN
NaN	45000

Strategies

1.1 Dropping Missing Values

Remove rows or columns containing missing values.

Example

Before		After Dropping Rows	
Age	Salary	Age	Salary
25	50000	25	50000
30	NaN		
40	70000	40	70000

Advantages

- Simple. No assumptions

Disadvantages

- Loss of information

1.2 Mean Imputation

Replace missing numerical values with the column mean.

Example

Scores	New values
80	80
90	90
NaN	80
70	70
Mean: $(80 + 90 + 70)/3 = 80$	

Best when

- Data is approximately normally distributed
- Few missing values

1.3 Median Imputation

Replace missing values with the median (middle value).

Example

10, 12, 13, 100, NaN

Median: 13

Median is less affected by outliers.

Best for

- Skewed distributions
- Data containing outliers

1.4 Mode Imputation

Replace missing values with the most frequent value.

Example

Color

- Red
- Blue
- Red
- Green
- NaN

Mode: Red

Best for Categorical variables

2. Detecting and Handling Outliers

An **outlier** is an observation that differs significantly from the rest of the data.

Example

20

21

22

23

24

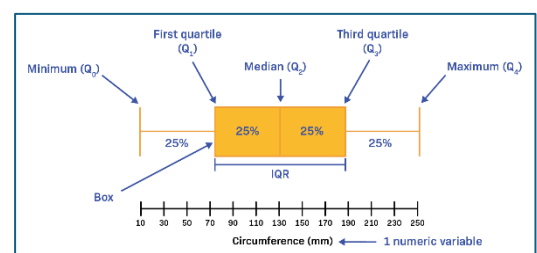
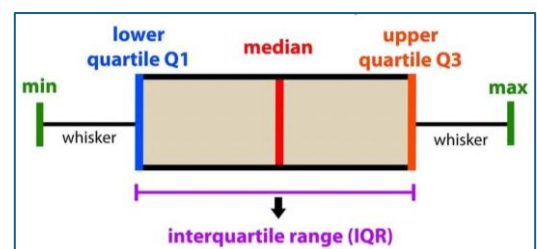
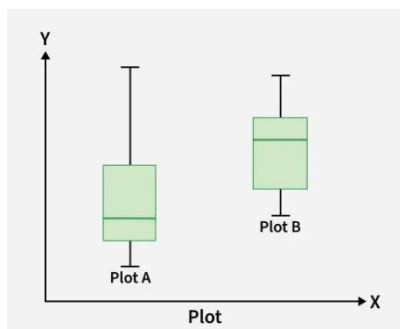
200

Here, 200 is an outlier.

Detecting Outliers

2.1 Box Plot

Points beyond the whiskers may indicate outliers.



2.2 Interquartile Range (IQR)

Calculate

- $IQR = Q3 - Q1$

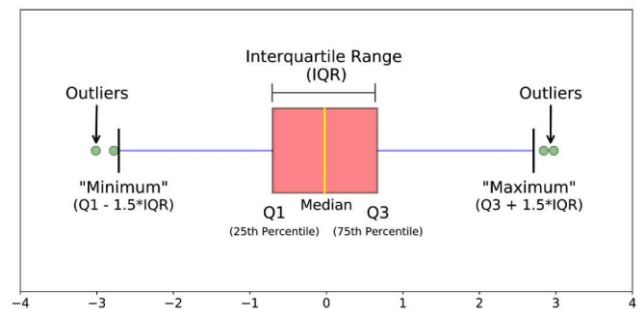
Lower Bound

- $Q1 - 1.5 \times IQR$

Upper Bound

- $Q3 + 1.5 \times IQR$

Values outside these bounds are considered outliers.

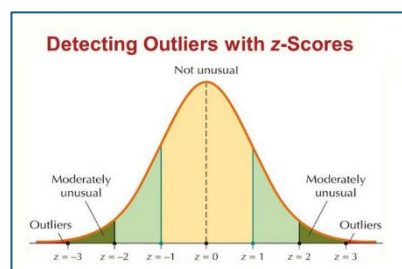


2.3 Z-Score

Measures how many standard deviations a value is from the mean.

Common threshold

$$|Z| > 3$$



Handling Outliers

- Remove them if they are data errors.
- Cap (Winsorize) extreme values.
- Apply transformations (e.g., log transformation).
- Use robust models or robust **scaling**.
- Leave them unchanged if they represent valid rare events.

3. Standardizing Inconsistent Text Data

Real-world text data often contains inconsistent formatting.

Example

USA

usa

U.S.A.

United States

These all refer to the same country.

After standardization

USA

USA

USA

USA

Common techniques

- Convert to lowercase or uppercase
- Remove extra spaces
- Remove punctuation
- Correct spelling
- Replace abbreviations
- Standardize date formats

Example

Before

- Male
- male
- MALE
- Male

After

- Male
- Male
- Male
- Male

Benefits

- Improves consistency
- Reduces duplicate categories
- Enables accurate grouping and analysis

4. Data Transformation Techniques

Data transformation changes data into a more suitable format for analysis or modeling.

Examples include

- Log transformation
- Square root transformation
- Reciprocal transformation
- Power transformation
- Binning
- **Scaling**
- **Encoding**

Example

Salary

- 1000
- 2000
- 100000

After log transformation

- 6.9
- 7.6
- 11.5

The distribution becomes less skewed.

5. Normalization and Standardization

These are scaling techniques used to make features comparable.

5.1. Normalization (Min-Max Scaling)

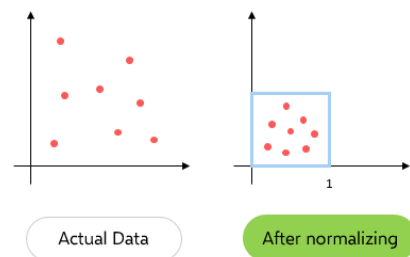
Scales values into a fixed range, usually **0 to 1**.

Formula

$$(x - \min) / (\max - \min)$$

Example

Original	Normalized
20	0
40	0.33
60	0.67
80	1



Best for

- Neural Networks
- KNN
- Distance-based algorithms

5.2 Standardization (Z-Score Scaling)

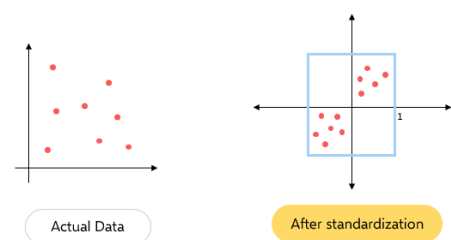
Centers data around a mean of **0** with a standard deviation of **1**.

Formula

$$(x - \text{mean}) / \text{standard_deviation}$$

Example

Original Age	Standardized
20	-1.34
30	-0.45
40	0.45
50	1.34



Best for

- Linear Regression
- Logistic Regression
- SVM
- PCA

6. Categorical Data Encoding

Machine learning algorithms require numerical input.

Encoding converts categories into numbers.

6.1 Label Encoding

Assign each category a unique integer.

Example

Color	Encoded
Red	0
Blue	1
Green	2

Suitable for

- Ordinal data
- Tree-based models (though one-hot encoding is often also effective)

Problem: The numbers may imply an order that doesn't exist for nominal categories.

6.2 One-Hot Encoding

Create a binary column for each category.

Example

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Advantages

- No false ordering
- Widely used for nominal categories

Disadvantages

- Can create many columns for high-cardinality features

7. Data Aggregation

Aggregation summarizes data into meaningful statistics.

7.1 groupby()

Groups rows based on one or more columns **and** applies aggregate functions.

Example

Sales

Product	Sales
Laptop	100
Phone	50
Laptop	150

Grouped Results

Product	Total Sales
Laptop	250
Phone	50

Common aggregations

- Sum
- Mean
- Count
- Maximum
- Minimum
- Median

7.2 Pivot Tables

Pivot tables summarize data across two or more dimensions.

Example

Pivot Table

Region	Product	Sales
East	Laptop	100
East	Phone	50
West	Laptop	150

Region	Laptop	Phone
East	100	50
West	150	0

Benefits

- Fast summaries
- Cross-tabulation
- Business reporting
- Interactive analysis

8. Matplotlib Visualizations

Visualization helps identify patterns, trends, and anomalies.

1. Univariate Analysis examines **one variable at a time**. Its goal is to understand the distribution and characteristics of a single feature.

Questions answered:

- What is the average?
- What is the range?
- Is the data skewed?
- Are there outliers?
- How are values distributed?

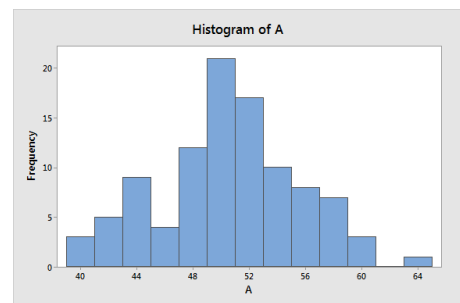
Common Plots

Histogram: Shows frequency distribution.

Use for:

- Age
- Salary
- Marks
- Sales

```
plt.hist(df["Age"])
```

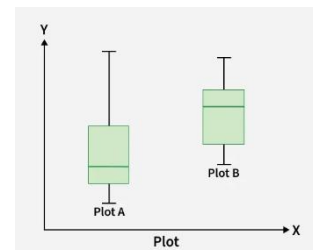


Box Plot

Shows

- Median
- Quartiles
- Outliers

```
plt.boxplot(df["Salary"])
```

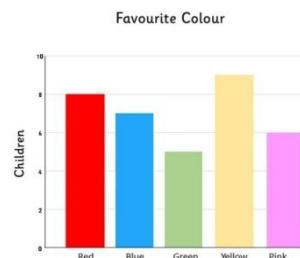


Count Plot (Bar Chart): For categorical variables.

Department

- HR
- IT
- Sales
- Finance

```
df["Department"].value_counts().plot(kind="bar")
```



Example Business Questions

- What is the average salary?
- Which department has the most employees?
- Are there salary outliers?

2. Bivariate Analysis: Studies the relationship between **two variables**.

Questions answered

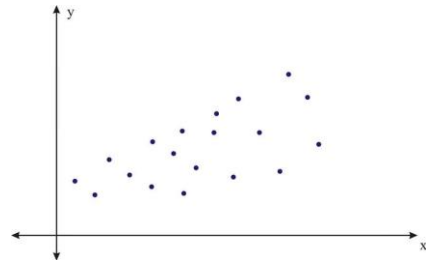
- Does X affect Y?
- Are two variables correlated?
- How do categories compare?

Numerical vs Numerical

Scatter Plot

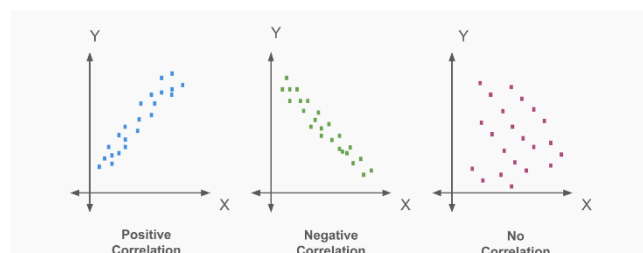
Salary vs Experience

```
plt.scatter(df["Experience"], df["Salary"])
```



Can identify

- Positive correlation
- Negative correlation
- Clusters
- Outliers



Numerical vs Categorical

Bar Chart

Average salary by department

```
df.groupby("Department")["Salary"].mean().plot(kind="bar")
```

Box Plot

Salary distribution by department

```
import seaborn as sns
sns.boxplot(x="Department", y="Salary", data=df)
```

Categorical vs Categorical

Grouped Bar Chart

Example: Department vs Gender

Example Business Questions

- Does experience increase salary?
- Which department pays more?
- Which category performs best?

3. Multivariate Analysis: Studies relationships among **three or more variables**.

Used for discovering complex patterns.

Bubble Plot

Example

- Experience
- Salary
- Age

```
plt.scatter(df["Experience"],df["Salary"],s=df["Age"]*10)
```

Bubble size -> Age

Scatter Plot with Color

```
plt.scatter(df["Experience"],df["Salary"],c=df["DepartmentCode"])
```

Heatmap (Correlation Matrix)

One of the most common multivariate plots.

```
sns.heatmap(df.corr(), annot=True)
```

Shows

- Strong correlations
- Weak correlations

Pair Plot

Every variable against every other variable.

```
sns.pairplot(df)
```

Useful during EDA.

Example Business Questions

- Which variables are highly correlated?
- Which features affect sales together?
- Which variables should be removed?

Summary Table

Analysis	Number of Variables	Common Charts	Purpose
Univariate	1	Histogram, Box Plot, Bar Chart	Understand one variable
Bivariate	2	Scatter Plot, Bar Chart, Box Plot	Study relationships
Multivariate	3+	Heatmap, Pair Plot, Bubble Plot	Analyze complex relationships

End of Document